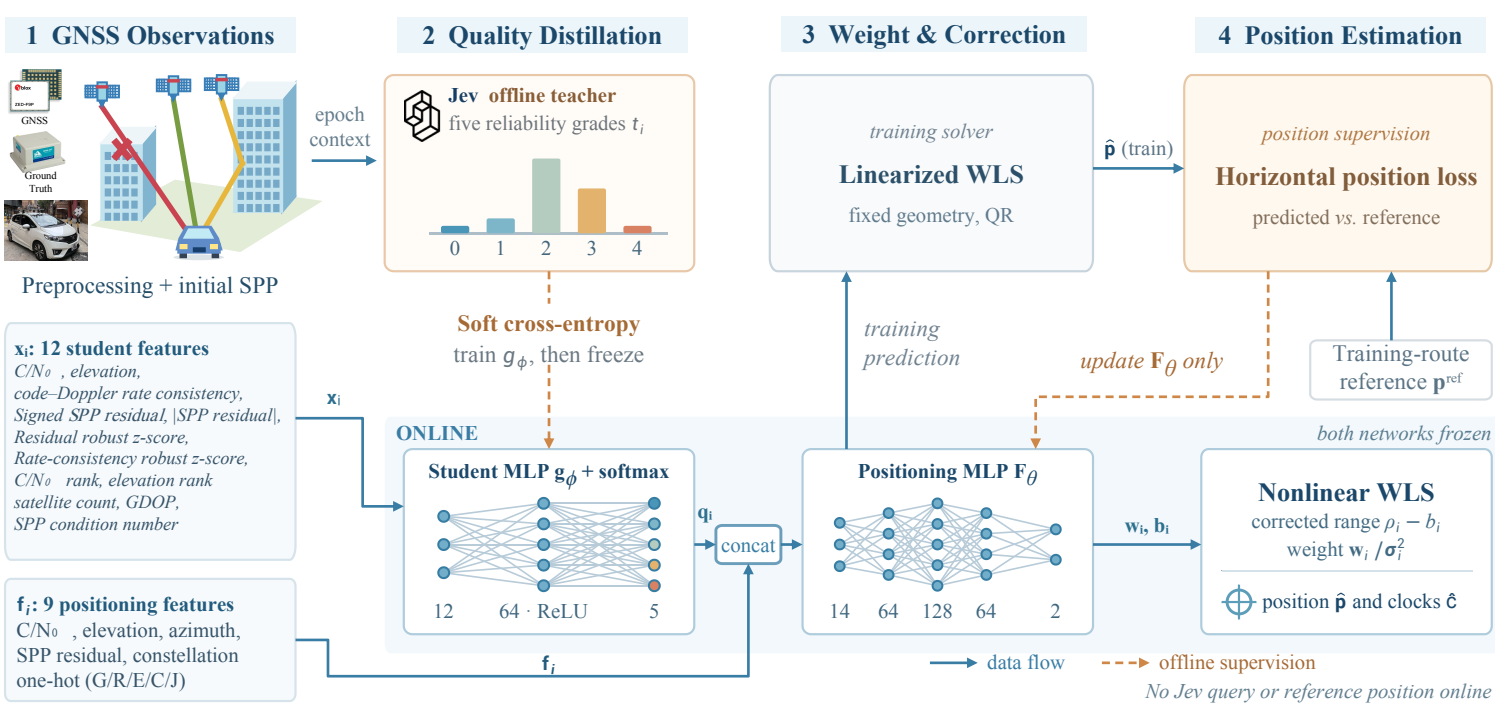

1 GNSS Observations
2 Quality Distillation
3 Weight & Correction
4 Position Estimation
GNSS
Ground Truth
Preprocessing + initial SPP
epoch context
Jev offline teacher
five reliability grades t_i
0 1 2 3 4
Soft cross-entropy
train g_ϕ, then freeze
training solver
Linearized WLS
fixed geometry, QR
$\hat{\mathbf{p}}$ (train)
position supervision
Horizontal position loss
predicted vs. reference
training prediction
update $\mathbf{F}_\theta$ only
Training-route reference $\mathbf{p}^{ref}$
$\mathbf{x}_i$: 12 student features
C/N_0 , elevation,
code–Doppler rate consistency,
Signed SPP residual, |SPP residual|,
Residual robust z-score,
Rate-consistency robust z-score,
C/N_0 rank, elevation rank
satellite count, GDOP,
SPP condition number
$\mathbf{x}_i$
ONLINE
both networks frozen
Student MLP $\mathbf{g}_\phi$ + softmax
12 64 · ReLU 5
$\mathbf{q}_i$
concat
Positioning MLP $\mathbf{F}_\theta$
14 64 128 64 2
$\mathbf{w}_i$, $\mathbf{b}_i$
Nonlinear WLS
corrected range $\rho_i - b_i$
weight $\mathbf{w}_i$ / σ_i^2
position $\hat{\mathbf{p}}$ and clocks $\hat{\mathbf{c}}$
$\mathbf{f}_i$: 9 positioning features
C/N_0 , elevation, azimuth,
SPP residual, constellation
one-hot (G/R/E/C/J)
$\mathbf{f}_i$
data flow
offline supervision
No Jev query or reference position online